Lecture 1: detailed solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to `Listas/lista-reais-lecture-01.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 1 — Static games I: dominance and iterated elimination

Item 1 (item 1 of the combined list)**: Tadelis Exercise 4.5** Foundations · difficulty 1/5

Source: Tadelis Exercise 4.5, page 77. The statement is reproduced verbatim in the list (item 1); it is not repeated here.

(a) What is asked. Which strategy profiles of the 3×3 game survive IESDS?

(b) Method. Strict dominance (Lecture 1, slide 23; Tadelis §4.1, pp. 59–61) and IESDS round by round (Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). In each round k a strategy of player i is deleted if some other surviving strategy of i gives a strictly higher payoff against *every* surviving strategy of the opponent; all such strategies are deleted at once. The procedure stops when a round deletes nothing. A best-response marking is added at the end to show how the survivors relate to Nash equilibrium (Lecture 2, slides 14–17).

(c) Solution. Matrix as printed in the book (p. 77):

$1 \setminus 2$	L	C	R
U	6, 8	2, 6	8, 2
M	8, 2	4, 4	9, 5
D	8, 10	4, 6	6, 7

Write each strategy as its payoff vector against the opponent's strategies: player 1's rows over columns (L, C, R) are $U = (6, 2, 8)$, $M = (8, 4, 9)$, $D = (8, 4, 6)$; player 2's columns over rows (U, M, D) are $L = (8, 2, 10)$, $C = (6, 4, 6)$, $R = (2, 5, 7)$.

Round 1 (S^0 = the whole game). Every pair of strategies is compared.

Player	Pair	Componentwise comparison	Verdict
1	U vs M	$6 < 8$, $2 < 4$, $8 < 9$	U strictly dominated by M
1	U vs D	$6 < 8$, $2 < 4$, $8 > 6$	not ranked
1	M vs D	$8 = 8$, $4 = 4$, $9 > 6$	D only weakly dominated by M
2	L vs C	$8 > 6$, $2 < 4$, $10 > 6$	not ranked
2	L vs R	$8 > 2$, $2 < 5$, $10 > 7$	not ranked
2	C vs R	$6 > 2$, $4 < 5$, $6 < 7$	not ranked

Only U is deleted: $S_1^1 = \{M, D\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 now compares columns on the surviving rows (M, D) only: $L = (2, 10)$, $C = (4, 6)$, $R = (5, 7)$.

- C vs R: $4 < 5$ and $6 < 7$, so C is strictly dominated by R.
- L vs C: $2 < 4$ but $10 > 6$; L vs R: $2 < 5$ but $10 > 7$. Not ranked.

Player 1 still faces all three columns: $M = (8, 4, 9)$ and $D = (8, 4, 6)$ tie against L and C, so neither strictly dominates the other. $S_1^2 = \{M, D\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $M = (8, 9)$, $D = (8, 6)$, a tie against L, so no strict dominance. Player 2 on (M, D): $L = (2, 10)$, $R = (5, 7)$, not ranked. Nothing is deleted and the process stops:

$$S^\infty = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Why no mixed strategy deletes more. In the final game each surviving strategy is a best reply to some surviving strategy of the opponent: L is the unique best reply to D ($10 > 7$), R the unique best reply to M ($5 > 2$), M the unique best reply to R ($9 > 6$), and D a best reply to L (tie at 8). A strategy that is a best reply to some belief cannot be strictly dominated by any pure or mixed strategy, so the answer does not depend on whether mixtures are allowed as dominators.

Best-response marking (underlined: the payoff is a best reply of its player to the other's strategy).

$1 \setminus 2$	L	C	R
U	6, <u>8</u>	2, 6	8, 2
M	<u>8</u>, 2	<u>4</u>, 4	<u>9</u>, <u>5</u>
D	<u>8</u>, <u>10</u>	<u>4</u>, 6	6, 7

Player 1: $BR_1(L) = \{M, D\}$, $BR_1(C) = \{M, D\}$, $BR_1(R) = \{M\}$. Player 2: $BR_2(U) = \{L\}$, $BR_2(M) = \{R\}$, $BR_2(D) = \{L\}$. Both payoffs are underlined in exactly two cells, (M, R) and (D, L): these are the pure Nash equilibria, and both lie in S^∞, as Lecture 2's Proposition 1 (slide 16) requires.

(d) Interpretation. Round 1 uses only player 1's rationality: U is worse than M whatever player 2 does. Round 2 uses player 2's knowledge that player 1 is rational: C is a sensible reply only to beliefs that put weight on U, and once U is gone R beats it everywhere (Lecture 2, slide 7). After that, no further layer of knowledge helps. The iterated-elimination equilibrium is a set of four profiles, not a point; Nash equilibrium, which adds correct beliefs, cuts it to (M, R) with payoffs $(9, 5)$ and (D, L) with $(8, 10)$. Deleting the weakly dominated D would single out (M, R), but weak dominance is not part of IESDS and its outcome can depend on the order of deletion.

(e) Verification. `check_A1.py` runs IESDS with the lecture's definition and finds exactly these rounds (round 1: U; round 2: C; then stop), that U is dominated only by M and C only by R, that D is weakly dominated by M in the final set, and, by brute force over the nine cells, that the pure Nash equilibria are (M, R) and (D, L). Tadelis gives no answer.

(f) Common mistakes.

- Deleting D because "M is at least as good everywhere". The ties against L and C make that weak dominance, which IESDS does not use.
- Deleting C in round 1. Against U it beats R ($6 > 2$); it becomes dominated only after U is gone, which is why the order of rounds matters for the explanation.
- Reporting the answer as one profile. The question asks for the surviving profiles: all four in $\{M, D\} \times \{L, R\}$.

Answer. Round 1: delete U (dominated by M). Round 2: delete C (dominated by R once U is gone). Nothing else is strictly dominated: $S^\infty = \{M, D\} \times \{L, R\}$, four profiles. (Of these, (M, R) and (D, L) are the pure Nash equilibria.)

Item 2 (item 2 of the combined list)**: Muñoz-Garcia Ch. 1, Exercise 13** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 1, Exercise 13, page 26. The statement is reproduced verbatim in the list (item 2); it is not repeated here.

(a) What is asked. For the 3×2 game of Fig. 1.33, find strictly dominant and strictly dominated strategies, then run IDSDS and explain each step.

(b) Method. Strictly dominant and strictly dominated strategies (Tadelis Defs. 4.1–4.2, pp. 60–61; Lecture 1, slide 23) and iterated deletion (Muñoz-Garcia's "IDSDS" is the slides' IESDS: Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). The point of the exercise is the difference between strict and weak dominance: only strict dominance is used.

(c) Solution.

$1 \setminus 2$	L	R
U	$-10,\ 10$	$0,\ 12$
M	$-12,\ 0$	$2,\ 2$
D	$-9,\ 0$	$1,\ 0$

(U, M, D, L, R abbreviate Up, Middle, Down, Left, Right.) Payoff vectors: player 1's rows over (L, R) are $U = (-10, 0)$, $M = (-12, 2)$, $D = (-9, 1)$; player 2's columns over (U, M, D) are $L = (10, 0, 0)$, $R = (12, 2, 0)$.

(a) Strictly dominant strategies. A strictly dominant strategy must beat every other strategy of the player against every opponent strategy.

- Player 1: against L the best row is D ($-9 > -10 > -12$); against R it is M ($2 > 1 > 0$). No row is best against both columns, so player 1 has no strictly dominant strategy.
- Player 2: R beats L in row U ($12 > 10$) and row M ($2 > 0$) but ties in row D ($0 = 0$). So R is weakly, not strictly, dominant. Player 2 has no strictly dominant strategy.

(b) Strictly dominated strategies. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	U vs M	$-10 > -12$, $0 < 2$	not ranked
1	U vs D	$-10 < -9$, $0 < 1$	U strictly dominated by D
1	M vs D	$-12 < -9$, $2 > 1$	not ranked
2	L vs R	$10 < 12$, $0 < 2$, $0 = 0$	L only weakly dominated by R

The only strictly dominated strategy is player 1's U (by D). Player 2 has none.

(c) IDSDS.

- *Step 1.* Delete U: player 1 is rational and D pays more than U whatever player 2 does.
- *Step 2.* In the reduced game (rows M, D), player 2 knows player 1 is rational, so she compares columns on rows (M, D) only: $L = (0, 0)$, $R = (2, 0)$. R is better in row M and ties in row D: weak dominance only, so nothing is deleted. Player 1 on columns (L, R): $M = (-12, 2)$, $D = (-9, 1)$, not ranked.

Nothing else can be deleted, and the surviving pairs are

$$\text{IDSDS} = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Beyond what is asked: best-response marking and Nash equilibria.

$1 \setminus 2$	L	R
U	$-10,\ 10$	$0,\ \underline{12}$
M	$-12,\ 0$	$\underline{2},\ \underline{2}$
D	$\underline{-9},\ \underline{0}$	$1,\ \underline{0}$

$BR_1(L) = \{D\}$, $BR_1(R) = \{M\}$; $BR_2(U) = \{R\}$, $BR_2(M) = \{R\}$, $BR_2(D) = \{L, R\}$. The pure Nash equilibria are (M, R), payoffs $(2, 2)$, and (D, L), payoffs $(-9, 0)$. The second one uses player 2's weakly dominated L, which is why it survives IDSDS but would be removed by weak dominance.

(d) Interpretation. Rationality alone removes U. A second layer of reasoning does not help, because player 2 is indifferent between L and R exactly when player 1 plays D, and D survives. The prediction is the four profiles $\{M, D\} \times \{L, R\}$. If player 2 broke ties toward R (weak dominance), player 1 would play M and the outcome would be (M, R); that is a stronger assumption than IDSDS makes. The book's own remark makes the same point: if player 2's payoff at (D, R) were 1 instead of 0, R would be strictly dominant.

(e) Verification. `check_A1.py` checks every pairwise comparison in the table, that IESDS deletes only U (round 1) and then stops at $\{M, D\} \times \{L, R\}$, that R weakly dominates L, and that the pure Nash equilibria are (M, R) and (D, L). *Comparison with the book's answer* (pp. 26–28): it agrees on all three parts (no strictly dominant strategy; only U strictly dominated, by D; IDSDS $= \{(M, L), (M, R), (D, L), (D, R)\}$). One slip: part (c) opens "As shown in part (a) of the exercise, strategy U is strictly dominated", but that was shown in part (b).

(f) Common mistakes.

- Deleting L because R is "always at least as good". The tie in row D makes it weak dominance, which IDSDS does not use.
- Calling D strictly dominant for player 1 because it dominates U. A dominant strategy must beat *every* other strategy, and M beats D against R.
- Giving the answer as payoffs $((2, 2), \dots)$ instead of strategy pairs.

Answer. (a) Neither player has a strictly dominant strategy (R is only weakly dominant for player 2). (b) Only player 1's U is strictly dominated (by D); player 2 has no strictly dominated strategy. (c) Delete U; then nothing else is strictly dominated: $\{(M, L), (M, R), (D, L), (D, R)\}$ survive.

Item 3 (item 3 of the combined list)**: Gibbons Problem 1.2** Foundations · difficulty 2/5

Source: Gibbons Problem 1.2, page 48. The statement is reproduced verbatim in the list (item 3); it is not repeated here.

(a) What is asked. Which strategies of the 3×3 game survive IESDS, and which profiles are pure-strategy Nash equilibria?

(b) Method. IESDS (Gibbons §1.1.B, pp. 4–7; Lecture 1, slides 28–29) and pure-strategy Nash equilibrium (Gibbons §1.1.C, pp. 7–12; Lecture 2, slide 14), found by marking best responses (Lecture 2, slide 10). Gibbons's Proposition B (p. 12; Lecture 2, Proposition 1, slide 16): every Nash equilibrium survives IESDS, so after the elimination only the surviving cells need to be searched. The full marking below confirms it independently.

(c) Solution. The matrix as printed (the top-left cell reads "2.0" in the scan; it is $(2, 0)$):

$1 \setminus 2$	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Payoff vectors: player 1's rows over (L, C, R) are $T = (2, 1, 4)$, $M = (3, 1, 2)$, $B = (1, 0, 3)$; player 2's columns over (T, M, B) are $L = (0, 4, 3)$, $C = (1, 2, 2)$, $R = (2, 3, 0)$.

Round 1. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	T vs M	$2 < 3$, $1 = 1$, $4 > 2$	not ranked
1	T vs B	$2 > 1$, $1 > 0$, $4 > 3$	B strictly dominated by T
1	M vs B	$3 > 1$, $1 > 0$, $2 < 3$	not ranked
2	L vs C	$0 < 1$, $4 > 2$, $3 > 2$	not ranked
2	L vs R	$0 < 2$, $4 > 3$, $3 > 0$	not ranked
2	C vs R	$1 < 2$, $2 < 3$, $2 > 0$	not ranked

C is *not* dominated yet: row B is what keeps it alive ($2 > 0$ against R). $S_1^1 = \{T, M\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 on rows (T, M): $L = (0, 4)$, $C = (1, 2)$, $R = (2, 3)$. C vs R: $1 < 2$ and $2 < 3$, so C is strictly dominated by R. L vs C ($0 < 1$, $4 > 2$) and L vs R ($0 < 2$, $4 > 3$) are not ranked. Player 1 on all three columns: T vs M unchanged, not ranked. $S_1^2 = \{T, M\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $T = (2, 4)$, $M = (3, 2)$, not ranked. Player 2 on (T, M): $L = (0, 4)$, $R = (2, 3)$, not ranked. Stop:

$$S^\infty = \{T, M\} \times \{L, R\}.$$

Pure Nash equilibria: best-response marking of the whole matrix.

$1 \setminus 2$	L	C	R
T	2, 0	$\underline{1}$, 1	$\underline{4}$, $\underline{2}$
M	$\underline{3}$, $\underline{4}$	$\underline{1}$, 2	2, 3
B	1, $\underline{3}$	0, 2	3, 0

Player 1: $BR_1(L) = \{M\}$ $(3 > 2 > 1)$, $BR_1(C) = \{T, M\}$ (tie at 1), $BR_1(R) = \{T\}$ $(4 > 3 > 2)$. Player 2: $BR_2(T) = \{R\}$ $(2 > 1 > 0)$, $BR_2(M) = \{L\}$ $(4 > 3 > 2)$, $BR_2(B) = \{L\}$ $(3 > 2 > 0)$. Both payoffs are underlined only in (T, R) and (M, L). Cell by cell inside S^∞:

- (T, L): player 1 would switch to M $(3 > 2)$. Not NE.
- (T, R): $T \in BR_1(R)$ and $R \in BR_2(T)$. NE, payoffs $(4, 2)$.
- (M, L): $M \in BR_1(L)$ and $L \in BR_2(M)$. NE, payoffs $(3, 4)$.
- (M, R): player 1 would switch to T $(4 > 2)$. Not NE.

(d) Interpretation. C is a sensible reply only to beliefs that put weight on B; once player 2 knows that player 1 is rational, C goes (Lecture 2, slide 7). IESDS stops at four profiles because each surviving strategy is a best reply to one of the opponent's surviving strategies. Nash equilibrium adds the requirement that beliefs be correct and cuts the four to two. The two equilibria are ranked differently by the players ((T, R) favours player 1, (M, L) player 2), so the game has a coordination problem that neither IESDS nor Nash equilibrium resolves.

(e) Verification. `check_A1.py` reproduces the rounds (round 1: B; round 2: C), checks that C is not dominated in the full game, and finds by brute force over all nine cells that the pure Nash equilibria are exactly (T, R) and (M, L). Gibbons gives no answer. Kumar's unofficial Gibbons manual (Answer 1.2, not in `Livros/`) gives the same deletions and equilibria; it writes the survivors as "(T,M) and (L,R)", meaning $\{T, M\}$ for player 1 and $\{L, R\}$ for player 2.

(f) Common mistakes.

- Deleting C in round 1 because it looks poor. In the full game C beats R in row B and L in row T; it is dominated only once B is gone.
- Reading "2.0" as the number two and leaving player 2's payoff blank. It is the pair $(2, 0)$; with any other reading the column comparisons change.
- Calling (T, L) or (M, R) an equilibrium because they survive IESDS. Survival is necessary for Nash equilibrium, not sufficient.

Answer. IESDS: delete B (by T), then C (by R); survivors $\{T, M\} \times \{L, R\}$. Pure Nash equilibria: (T, R) with payoffs $(4, 2)$ and (M, L) with payoffs $(3, 4)$.

Item 4 (item 4 of the combined list)**: Tadelis Exercise 4.1** Exam · difficulty 2/5

Source: Tadelis Exercise 4.1, page 76. The statement is reproduced verbatim in the list (item 4); it is not repeated here.

(a) What is asked. Prove that a strictly dominant strategy equilibrium, when it exists, is the only dominant strategy equilibrium.

(b) Method. Definitions only: strict dominance (Tadelis Def. 4.1, p. 60), strictly dominant strategy (Def. 4.2, p. 61), strict dominant strategy equilibrium (Def. 4.3, p. 61) and Proposition 4.1 itself (p. 61); in the slides, Lecture 1, slides 23 and 25. The proof is by contradiction, one player at a time. The statement says "the unique dominant strategy equilibrium" without the word "strict", so the proof also rules out a second profile of *weakly* dominant strategies (Lecture 1, slide 23), which is the stronger reading.

(c) Solution. Notation. $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$; for a profile s, s_{-i} is the profile of the others. Every S_j is non-empty, so S_{-i} is non-empty. s_i is *strictly dominant* if $v_i(s_i, s_{-i}) > v_i(s_i', s_{-i})$ for all $s_i' \neq s_i$ and all s_{-i}; it is *weakly dominant* if $v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i})$ for all s_i' and all s_{-i}, with strict inequality somewhere for each $s_i' \neq s_i$. A (strict, weak) dominant strategy equilibrium is a profile in which every player uses a (strictly, weakly) dominant strategy.

Step 1: a player has at most one dominant strategy when one of them is strictly dominant. Let s_i^D be strictly dominant for player i and let $t_i \neq s_i^D$. By Def. 4.2, for every $s_{-i} \in S_{-i}$,

$$v_i(s_i^D, s_{-i}) > v_i(t_i, s_{-i}). \tag{1}$$

Suppose t_i were also dominant, strictly or weakly. Then, in particular,

$$v_i(t_i, s_{-i}) \geq v_i(s_i^D, s_{-i}) \quad \text{for every } s_{-i} \in S_{-i}. \tag{2}$$

Pick any $\hat{s}_{-i} \in S_{-i}$ (possible because $S_{-i} \neq \emptyset$). Then (1) and (2) give $v_i(s_i^D, \hat{s}_{-i}) > v_i(t_i, \hat{s}_{-i}) \geq v_i(s_i^D, \hat{s}_{-i})$, a contradiction. So no $t_i \neq s_i^D$ is dominant in either sense.

Step 2: the proposition. Let s^D be a strict dominant strategy equilibrium, and suppose some profile $s' \neq s^D$ is also a dominant strategy equilibrium. Since $s' \neq s^D$, there is a player i with $s_i' \neq s_i^D$. In s', player i's strategy s_i' is dominant; but s_i^D is strictly dominant for i (because s^D is a strict dominant strategy equilibrium), so by Step 1 no strategy other than s_i^D can be dominant for i. This contradicts $s_i' \neq s_i^D$. Hence s^D is the unique dominant strategy equilibrium. ■

Two remarks that complete the picture.

- The proposition does not claim existence: Battle of the Sexes has no dominant strategy equilibrium (Tadelis p. 62).
- Uniqueness extends to the other solution concepts of the course: s^D is the unique profile that survives IESDS (Lecture 1, slide 29: in round 1 every $s_i' \neq s_i^D$ is strictly dominated by s_i^D) and hence the unique Nash equilibrium (Lecture 2, slide 17). Without strictness uniqueness can fail: if dominance only required "$\geq$ against every s_{-i}", a player whose payoffs are all equal would have every strategy dominant, and every profile of such players would be a dominant strategy equilibrium.

(d) Interpretation. A strictly dominant strategy is a best reply that does not depend on beliefs. Two such strategies for the same player would each have to do strictly better than the other against the same opponents, which is impossible. That is why the dominant-strategy prediction, when it exists, is a single profile, and why it needs only rationality, not common knowledge of it.

(e) Verification. `check_A1.py` draws 3,000 random finite games (two or three players, two to four strategies each, integer payoffs), plants a strictly dominant strategy for every player in half of them, and checks in each game that, whenever a strict dominant strategy equilibrium exists, no other profile consists of weakly or strictly dominant strategies, and that it is the only IESDS survivor. Tadelis gives no answer.

(f) Common mistakes.

- Proving only that there is no second *strict* dominant strategy equilibrium. The proposition says "dominant", and the same two-line argument rules out weakly dominant ones.
- Comparing the two profiles s^D and s' as a whole. Dominance is a property of one player's strategy against all of the others' profiles; the contradiction comes from a single player whose strategies differ.
- Forgetting that S_{-i} must be non-empty for the contradiction (one $\hat{s}_{-i}$ is needed). It always is in a game, but the step should be stated.

Answer. If s^D is a strict dominant strategy equilibrium and $s' \neq s^D$, some player has $s'_i \neq s_i^D$. Strict dominance of s_i^D gives $v_i(s_i^D, s_{-i}) > v_i(s'_i, s_{-i})$ for every s_{-i}, so s'_i cannot be dominant, strictly or weakly (that would need $v_i(s'_i, s_{-i}) \geq v_i(s_i^D, s_{-i})$ for every s_{-i}). Hence s' is not a dominant strategy equilibrium, and s^D is the unique one.

Item 5 (item 5 of the combined list)**: Muñoz-Garcia Ch. 1, Exercise 12**Exam · difficulty 2/5

Source: Muñoz-Garcia Ch. 1, Exercise 12, page 23. The statement is reproduced verbatim in the list (item 5); it is not repeated here.

(a) What is asked. For the 3×3 game of Fig. 1.30, find dominant and dominated strategies, then what is deleted under rationality alone and under common knowledge of rationality.

(b) Method. Strict dominance and dominant strategies (Lecture 1, slides 23 and 25; Tadelis §4.1, pp. 59–61) and the epistemic reading of IESDS (Lecture 2, slide 7; Tadelis §4.2.1, pp. 63–65). Round 1 of IESDS uses only each player's own rationality; round 2 adds "each knows the other is rational"; round k adds $k-1$ layers of that knowledge. Part (c) stops after round 1, part (d) runs IESDS to the end (Lecture 1, slide 29).

(c) Solution. Matrix from the book (p. 23):

$1 \setminus 2$	F	C	B
F	0, 5	2, 4	2, 3
C	2, 3	0, 5	3, 2
B	5, 0	2, 4	2, 3

Payoff vectors: player 1's rows over columns (F, C, B) are $F = (0, 2, 2)$, $C = (2, 0, 3)$, $B = (5, 2, 2)$; player 2's columns over rows (F, C, B) are $F = (5, 3, 0)$, $C = (4, 5, 4)$, $B = (3, 2, 3)$.

Best-response marking (used in (a) and at the end).

$1 \setminus 2$	F	C	B
F	0, $\underline{5}$	$\underline{2}$, 4	2, 3
C	2, 3	0, $\underline{5}$	$\underline{3}$, 2
B	$\underline{5}$, 0	$\underline{2}$, $\underline{4}$	2, 3

$BR_1(F) = \{B\}$, $BR_1(C) = \{F, B\}$, $BR_1(B) = \{C\}$; $BR_2(F) = \{F\}$, $BR_2(C) = \{C\}$, $BR_2(B) = \{C\}$.

(a) Strictly dominant strategies. A strictly dominant strategy would be the unique best reply to every strategy of the opponent. Player 1's best reply changes with the column (B, then $\{F, B\}$, then C), and player 2's changes with the row (F, then C). Neither player has a strictly dominant strategy.

(b) Strictly dominated strategies. All pairs:

Player	Pair	Componentwise comparison	Verdict
1	F vs C	$0 < 2$, $2 > 0$, $2 < 3$	not ranked
1	F vs B	$0 < 5$, $2 = 2$, $2 = 2$	F only weakly dominated by B
1	C vs B	$2 < 5$, $0 < 2$, $3 > 2$	not ranked
2	F vs C	$5 > 4$, $3 < 5$, $0 < 4$	not ranked
2	F vs B	$5 > 3$, $3 > 2$, $0 < 3$	not ranked
2	C vs B	$4 > 3$, $5 > 2$, $4 > 3$	B strictly dominated by C

No mixed strategy dominates anything else either: F, C, B of player 1 and F, C of player 2 are each a best reply to some opponent strategy (marking above). The only strictly dominated strategy is player 2's B.

(c) Rationality, no common knowledge. Each player deletes his own strictly dominated strategies but does not use the fact that the other does the same. Player 2 drops B; player 1 has nothing to drop and still evaluates his rows against all three columns. The surviving profiles are

$$\{F, C, B\} \times \{F, C\} = \{(F, F), (F, C), (C, F), (C, C), (B, F), (B, C)\}.$$

(d) Common knowledge of rationality.

- Round 2: player 1 knows player 2 is rational, so he compares rows on columns $\{F, C\}$ only: $F = (0, 2)$, $C = (2, 0)$, $B = (5, 2)$. C vs B: $2 < 5$, $0 < 2$, so C is strictly dominated by B. F vs B: $0 < 5$, $2 = 2$, only weak. F vs C: not ranked. Player 2 still faces all three rows; F vs C is not ranked. $S_1^2 = \{F, B\}$, $S_2^2 = \{F, C\}$.
- Round 3: player 2 on rows (F, B): $F = (5, 0)$, $C = (4, 4)$, not ranked. Player 1 on columns (F, C): $F = (0, 2)$, $B = (5, 2)$, only weak. Stop.

$$S^\infty = \{F, B\} \times \{F, C\} = \{(F, F), (F, C), (B, F), (B, C)\}.$$

Beyond what is asked. In the marking, both payoffs are underlined only in (B, C): it is the unique pure-strategy Nash equilibrium, with payoffs $(2, 4)$, and it lies in S^∞.

(d) Interpretation. Rationality alone removes only strategies that are bad against *everything*. Player 1's C is the best reply to player 2's B, so it can be discarded only by someone who knows that player 2 will not play B, that is, by someone who uses player 2's rationality. Each extra layer of "I know that you know" can remove more strategies; here the second layer removes C and the third removes nothing. The difference between (c) and (d) is the difference between S^1 and S^∞ in the slides.

(e) Verification. `check_A1.py` checks the dominance statements in (a)–(b), that IESDS removes B for player 2 in round 1 and C for player 1 in round 2, that F is only weakly dominated by B, and that (B, C) is the only pure Nash equilibrium. The book's answer (pp. 23–26) agrees with (a)–(d). Its text for (b) and (d) describes the comparisons loosely (for example, it says player 1 deletes C because his payoff "is higher from B ... when player 2 plays F"), but the deletions and the final sets are the same as ours. The figure caption "three-players game" is a slip in the book: the game has two players.

(f) Common mistakes.

- Deleting player 1's F in (b) because B is never worse. The ties against C and B make it weak dominance only; F survives IESDS.
- In (c), letting player 1 compare rows only on $\{F, C\}$. That already uses player 2's rationality, which is the step (c) forbids.
- Thinking that "no dominant strategy" implies "no dominated strategy": (a) and (b) are separate questions, and player 2's B is dominated although no strategy is dominant.

Answer. (a) None. (b) Only player 2's B (strictly dominated by C). (c) Rationality alone: $\{F, C, B\} \times \{F, C\}$, six profiles. (d) With common knowledge, player 1 then drops C (dominated by B on $\{F, C\}$): $S^\infty = \{F, B\} \times \{F, C\}$, four profiles.

Item 6 (item 6 of the combined list)**: Tadelis Exercise 4.7** Exam · difficulty 3/5

Source: Tadelis Exercise 4.7, page 77–78. The statement is reproduced verbatim in the list (item 6); it is not repeated here.

(a) What is asked. Turn the campaigning story into a normal-form game and matrix, then run IESDS and say whether it gives a clear prediction.

(b) Method. Normal-form game $G = (I, \{S_i\}, \{u_i\})$ (Lecture 1, slides 13–14; Tadelis §3.1, pp. 46–50); IESDS (Lecture 1, slides 28–29; Tadelis §4.2, pp. 63–65). The only modelling step is to fill the payoffs the story leaves implicit, using the fact that exactly one candidate wins.

(c) Solution. **(a) The game.** Players $I = \{1, 2\}$; strategies $S_1 = S_2 = \{P, B, N\}$. Let $\pi(s_i, s_j)$ be the probability that the candidate playing s_i wins against one playing s_j. The book gives four rules:

$$\pi(s, s) = 0.5, \qquad \pi(P, B) = 0, \qquad \pi(P, N) = 0.3, \qquad \pi(N, B) = 0.6.$$

Exactly one candidate wins, so $\pi(s_i, s_j) + \pi(s_j, s_i) = 1$. This fills the other three entries:

$$\pi(B, P) = 1 - \pi(P, B) = 1, \qquad \pi(N, P) = 1 - 0.3 = 0.7, \qquad \pi(B, N) = 1 - 0.6 = 0.4.$$

Payoffs: $u_1(s_1, s_2) = \pi(s_1, s_2)$ and $u_2(s_1, s_2) = \pi(s_2, s_1) = 1 - u_1(s_1, s_2)$. Player 2's payoff function is the same rule π with the roles swapped (the game is symmetric), and it is the complement of player 1's because win probabilities add up to one (the game is constant-sum).

(b) The matrix, with best responses underlined:

$1 \setminus 2$	P	B	N
P	0.5, 0.5	0, $\underline{1}$	0.3, $\underline{0.7}$
B	$\underline{1}$, 0	0.5, 0.5	0.4, $\underline{0.6}$
N	0.7, 0.3	$\underline{0.6}$, 0.4	$\underline{0.5}$, $\underline{0.5}$

$BR_1(P) = \{B\}$, $BR_1(B) = \{N\}$, $BR_1(N) = \{N\}$, and by symmetry $BR_2(P) = \{B\}$, $BR_2(B) = \{N\}$, $BR_2(N) = \{N\}$.

(c) IESDS. Player 1's rows over columns (P, B, N): $P = (0.5, 0, 0.3)$, $B = (1, 0.5, 0.4)$, $N = (0.7, 0.6, 0.5)$.

- *Round 1.* P vs B: $0.5 < 1$, $0 < 0.5$, $0.3 < 0.4$, so P is strictly dominated by B. P vs N: $0.5 < 0.7$, $0 < 0.6$, $0.3 < 0.5$, so P is also dominated by N. B vs N: $1 > 0.7$ but $0.5 < 0.6$, not ranked. By symmetry player 2's columns satisfy the same comparisons. $S_i^1 = \{B, N\}$ for $i = 1, 2$.
- *Round 2.* On the columns $\{B, N\}$: $B = (0.5, 0.4)$, $N = (0.6, 0.5)$. $0.5 < 0.6$ and $0.4 < 0.5$, so B is strictly dominated by N. The same for player 2. $S_i^2 = \{N\}$.
- *Round 3.* One strategy each: nothing to compare.

$S^\infty = \{(N, N)\}$: both run negative campaigns and each wins with probability 0.5. The procedure gives a clear prediction, a unique profile, after two rounds. In the marking (N, N) is the only cell with both payoffs underlined, so it is also the unique Nash equilibrium (as it must be: Lecture 2, Proposition 2, slide 17).

(d) Interpretation. B survives round 1 only because it crushes a positive campaign. Once each candidate knows the other will not run a purely positive campaign, going fully negative wins. The outcome has the same win probabilities as (P, P), and in a constant-sum game every profile is Pareto optimal *for the candidates*. If voters dislike negative campaigning, the equilibrium is a prisoner's-dilemma outcome for them: the candidates end up exactly where they would have been with positive campaigns, but with a nastier campaign.

(e) Verification. `check_A1.py` builds the matrix from the four rules plus complementarity, checks symmetry, that P is dominated by both B and N, that IESDS removes P (round 1) and B (round 2) for both players, leaving (N, N), and that (N, N) is the only pure Nash equilibrium. Tadelis gives no answer.

(f) Common mistakes.

- Filling the missing cells by symmetry of *payoffs* ($\pi(B, P) = \pi(P, B) = 0$) instead of by complementarity. A candidate who beats P for sure must win with probability 1.
- Deleting B in round 1. Against P it is the unique best reply ($1 > 0.7$); it becomes dominated only after P is gone.
- In (a), writing only the matrix. The question asks for players, strategy sets and payoff functions, and for why player 2's function differs ($u_2 = 1 - u_1$).

Answer. (a) $I = \{1, 2\}$, $S_i = \{P, B, N\}$, $u_1 = \pi(s_1, s_2)$ from the rules plus $\pi(s_i, s_j) + \pi(s_j, s_i) = 1$, and $u_2 = 1 - u_1$ (the same rule from candidate 2's side). (b) Matrix above. (c) Round 1 deletes P (dominated by B and by N); round 2 deletes B (dominated by N); unique prediction (N, N), each wins with probability 0.5.

Item 7 (item 7 of the combined list)**: Tadelis Exercise 4.3** Exam · difficulty 3/5

Source: Tadelis Exercise 4.3, page 76. The statement is reproduced verbatim in the list (item 7); it is not repeated here.

(a) What is asked. Write the two-bidder first-price auction with bids $\{0, 1, 2\}$ as a matrix, find the strictly dominated strategies, and run IESDS.

(b) Method. Building the normal form from the rules (Lecture 1, slides 13–19; Tadelis §3.1, pp. 46–50); strict dominance (Tadelis Def. 4.1, p. 60; Lecture 1, slide 23); IESDS with every dominated strategy removed in each round (Lecture 1, slide 29; Tadelis §4.2, pp. 63–65). Ties are broken by a fair coin, so a tie gives each bidder half of (value − bid): payoffs are expected payoffs over the coin toss.

(c) Solution. (a) The matrix. Player 1 values the item at 3, player 2 at 5; $b_i \in \{0, 1, 2\}$. Player i's payoff is $\theta_i - b_i$ if $b_i > b_j$, 0 if $b_i < b_j$, and $\frac{1}{2}(\theta_i - b_i)$ if $b_i = b_j$, with $\theta_1 = 3$, $\theta_2 = 5$. Cell by cell:

- ties: $(0,0)$: $(\frac{3}{2}, \frac{5}{2})$; $(1,1)$: $(\frac{3-1}{2}, \frac{5-1}{2}) = (1, 2)$; $(2,2)$: $(\frac{1}{2}, \frac{3}{2})$;
- player 1 wins: $(1,0)$: $(3-1, 0) = (2, 0)$; $(2,0)$ and $(2,1)$: $(3-2, 0) = (1, 0)$;
- player 2 wins: $(0,1)$: $(0, 5-1) = (0, 4)$; $(0,2)$ and $(1,2)$: $(0, 5-2) = (0, 3)$.

1 \ 2	0	1	2
0	1.5, 2.5	0, 4	0, 3
1	2, 0	1, 2	0, 3
2	1, 0	1, 0	0.5, 1.5

Payoff vectors: player 1's rows over $b_2 = (0, 1, 2)$ are $0 \mapsto (1.5, 0, 0)$, $1 \mapsto (2, 1, 0)$, $2 \mapsto (1, 1, 0.5)$; player 2's columns over $b_1 = (0, 1, 2)$ are $0 \mapsto (2.5, 0, 0)$, $1 \mapsto (4, 2, 0)$, $2 \mapsto (3, 3, 1.5)$.

(b) Strictly dominated strategies (by pure strategies). All pairs:

Player	Pair	Componentwise comparison	Verdict
1	0 vs 1	$1.5 < 2$, $0 < 1$, $0 = 0$	0 only weakly dominated by 1
1	0 vs 2	$1.5 > 1$, $0 < 1$, $0 < 0.5$	not ranked
1	1 vs 2	$2 > 1$, $1 = 1$, $0 < 0.5$	not ranked
2	0 vs 1	$2.5 < 4$, $0 < 2$, $0 = 0$	0 only weakly dominated by 1
2	0 vs 2	$2.5 < 3$, $0 < 3$, $0 < 1.5$	0 strictly dominated by 2
2	1 vs 2	$4 > 3$, $2 < 3$, $0 < 1.5$	not ranked

Yes: player 2's bid 0 is strictly dominated by bid 2. Player 1 has no strategy strictly dominated by another pure strategy. (*Remark, mixed dominators*, Tadelis §6.1 and Lecture 2, slide 42: player 1's bid 0 *is* strictly dominated by the mixed strategy $\frac{3}{4}\cdot 1 + \frac{1}{4}\cdot 2$, which pays $(1.75,\ 1,\ 0.125) > (1.5,\ 0,\ 0)$. With Chapter 4's pure-strategy definition this does not count, and it does not change the answer to (c).)

(c) IESDS.

- *Round 1.* Delete player 2's 0 (by 2). Nothing for player 1. $S_1^1 = \{0, 1, 2\}$, $S_2^1 = \{1, 2\}$.

- *Round 2.* Player 1 over $b_2 \in \{1,2\}$: $0 \mapsto (0,0)$, $1 \mapsto (1,0)$, $2 \mapsto (1,0.5)$. 0 vs 2: $0 < 1$, $0 < 0.5$, so 0 is strictly dominated by 2. 1 vs 2: $1 = 1$, only weak. Player 2 over $b_1 \in \{0,1,2\}$: $1 \mapsto (4,2,0)$, $2 \mapsto (3,3,1.5)$, not ranked. $S_1^2 = \{1,2\}$, $S_2^2 = \{1,2\}$.
- *Round 3.* Player 2 over $b_1 \in \{1,2\}$: $1 \mapsto (2,0)$, $2 \mapsto (3,1.5)$): 1 is strictly dominated by 2. Player 1 over $b_2 \in \{1,2\}$: $1 \mapsto (1,0)$, $2 \mapsto (1,0.5)$, only weak. $S_1^3 = \{1,2\}$, $S_2^3 = \{2\}$.
- *Round 4.* Player 1 against $b_2 = 2$ only: bid 1 pays 0, bid 2 pays 0.5, so 1 is strictly dominated by 2. $S_1^4 = \{2\}$, $S_2^4 = \{2\}$.

$S^\infty = \{(2,2)\}$: both bid 2, each wins with probability $\frac{1}{2}$, and the expected payoffs are $(0.5, 1.5)$. Since the survivor is unique, it is the unique Nash equilibrium (Lecture 2, slide 17); the best-response marking confirms it: $BR_1(2) = \{2\}$ and $BR_2(2) = \{2\}$, while every other cell has a profitable deviation.

(d) Interpretation. Each round adds one layer of "I know that you know": player 2 never bids 0 (her value is high enough that bidding 2 is always better); knowing that, player 1 never bids 0 either; then player 2 bids 2 to avoid losing ties at 1; then player 1 matches. The coarse bid grid matters: the high-value bidder wins only half the time, so the item goes to the bidder who values it less with probability $\frac{1}{2}$, an inefficiency created by the tie rule and the bid cap.

(e) Verification. `check_A1.py` builds the matrix from the auction rules in exact arithmetic, checks the dominance table, runs IESDS and finds the four rounds above (player 2's 0; player 1's 0; player 2's 1; player 1's 1), confirms that $(2,2)$ is the only pure Nash equilibrium, and checks that $\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 2$ strictly dominates player 1's bid 0. Tadelis gives no answer.

(f) Common mistakes.

- Giving the tie winner the whole surplus or splitting the item rather than the expected payoff. With a fair coin each tied bidder gets $\frac{1}{2}(\theta_i - b_i)$.
- Charging the loser his bid. In a first-price auction only the winner pays.
- Stopping after round 1 or 2. The deletions alternate between the players, and only the fourth round leaves a single profile.

Answer. (a) Matrix above (rows: player 1's bid; columns: player 2's bid). (b) Player 2's bid 0 is strictly dominated by bid 2; player 1 has no strictly dominated pure strategy (his bid 0 is only weakly dominated by 1). (c) IESDS deletes, in order, player 2's 0, player 1's 0, player 2's 1, player 1's 1; only $(2,2)$ survives, with expected payoffs $(0.5, 1.5)$.

Item 8 (item 8 of the combined list)**: Tadelis Exercise 4.6** Exam · difficulty 4/5

Source: Tadelis Exercise 4.6, page 77. The statement is reproduced verbatim in the list (item 8); it is not repeated here.

(a) What is asked. For cleaning times $t_i \geq 0$ with payoff $(10 - t_j)t_i - t_i^2$, find the best response, the survivors of one round of IESDS, and the survivors of IESDS.

(b) Method. Best response (Lecture 2, slide 10; Tadelis §4.3.1, p. 69) and IESDS with a continuum of strategies (Lecture 1, slide 29), done exactly as in Tadelis's Cournot example (§4.2.2, pp. 65–67, Fig. 4.1) and in the slides' Cournot remark (Lecture 2, slide 24): bound the set of possible best replies and show that every action outside it is strictly dominated by the nearest end of the range.

(c) Solution. **(a) Best response.** Fix $t_j \geq 0$ and maximise $u_i(t_i, t_j) = (10 - t_j)t_i - t_i^2$ over $t_i \geq 0$.

- $\partial u_i/\partial t_i = 10 - t_j - 2t_i$ and $\partial^2 u_i/\partial t_i^2 = -2 < 0$: u_i is strictly concave, so the first-order condition with the constraint $t_i \geq 0$ gives the unique maximiser.
- If $t_j < 10$, the FOC $10 - t_j - 2t_i = 0$ gives $t_i = (10 - t_j)/2 > 0$.
- If $t_j \geq 10$, $\partial u_i/\partial t_i = 10 - t_j - 2t_i < 0$ for every $t_i > 0$, so the maximum is at the corner $t_i = 0$.

$$BR_i(t_j) = \max\Big\{0,\ 5 - \frac{t_j}{2}\Big\}.$$

It is single-valued and decreasing in t_j: cleaning is a strategic substitute.

Dominance lemma. Write $x = (10 - t_j)/2$ for the unconstrained peak, so $u_i(t, t_j) = 2xt - t^2 = t(2x - t)$. For any two times a and t,

$$u_i(a, t_j) - u_i(t, t_j) = 2x(a - t) - (a^2 - t^2) = (a - t)(2x - a - t) = (t - a)(t + a - 2x). \quad (1)$$

If $t > a \geq x$, both factors on the right are positive; if $t < a \leq x$, both are negative. So: if every peak x that the opponent's surviving strategies can generate lies in $[\underline{a}, \bar{a}]$, then every $t > \bar{a}$ is strictly dominated by $\bar{a}$ and every $t < \underline{a}$ by $\underline{a}$. Conversely, a t that is the unique best reply to some surviving t_j is never strictly dominated.

(b) One round. With $t_j \in [0, \infty)$, the best replies range over $[0, 5]$ ($BR_i(0) = 5$, $BR_i(t_j) = 0$ for $t_j \geq 10$), and the peaks $x = 5 - t_j/2$ are all ≤ 5.

- Every $t_i > 5$ is strictly dominated by 5: by (1) with $a = 5 \geq x$, for every t_j (if $x < 0$ the inequality still holds because $t + a - 2x > 0$).
- Every $t_i \in [0, 5]$ is the unique best reply to $t_j = 10 - 2t_i \in [0, 10]$, so it is not dominated.

Hence $S_i^1 = [0, 5]$ for both roommates.

(c) Iterating. Suppose $S_j^k = [L_k, U_k]$. Because BR_i is continuous and decreasing, the best replies to $t_j \in [L_k, U_k]$ fill exactly $[BR_i(U_k), BR_i(L_k)]$; by (1) everything outside that interval is strictly dominated by its nearer endpoint, and everything inside is a best reply. So

$$L_{k+1} = 5 - \frac{U_k}{2}, \qquad U_{k+1} = 5 - \frac{L_k}{2}, \qquad (L_1, U_1) = (0, 5).$$

(The $\max\{0, \cdot\}$ never binds from round 2 on because $U_k \leq 5 < 10$.)

k	1	2	3	4	5	∞
L_k	0	$5-5/2=2.5$	2.5	$5-3.75/2=3.125$	3.125	10/3
U_k	5	$5-0/2=5$	$5-2.5/2=3.75$	3.75	$5-3.125/2=3.4375$	10/3

Convergence. Subtracting the two recursions, $U_{k+1}-L_{k+1}=\frac{1}{2}(U_k-L_k)$, so $U_k-L_k=5/2^{k-1}\to 0$. Adding them, the midpoint $M_k=(L_k+U_k)/2$ satisfies $M_{k+1}=5-M_k/2$. Its fixed point solves $M=5-M/2$, i.e. $M=10/3$, and

$$M_k-\tfrac{10}{3}=(-\tfrac{1}{2})^{k-1}(M_1-\tfrac{10}{3})=(-\tfrac{1}{2})^{k-1}(\tfrac{5}{2}-\tfrac{10}{3})=(-\tfrac{1}{2})^{k-1}(-\tfrac{5}{6})\to 0.$$

The intervals are nested, their widths go to zero and their midpoints go to 10/3, so $S_i^\infty=\bigcap_k S_i^k=\{10/3\}$ for both roommates. IESDS predicts the unique profile $t_1=t_2=10/3$. It is also the Nash equilibrium: $t=5-t/2$ gives $t=10/3$.

(d) Interpretation. Each round uses one more layer of knowledge: round 1, "I am rational" (never clean more than 5 hours); round 2, "I know you are rational" (you clean at most 5, so I clean at least 2.5); and so on (Lecture 2, slide 7). Each roommate gets $u_i=(10-10/3)(10/3)-(10/3)^2=200/9-100/9=100/9\approx 11.1$. Joint welfare $2[(10-t)t-t^2]=2(10t-2t^2)$ is maximised at $10-4t=0$, i.e. $t=2.5$, where each gets $(7.5)(2.5)-6.25=12.5>100/9$. The IESDS outcome over-cleans relative to the joint optimum: each roommate ignores that his cleaning lowers the value of the other's (a negative externality, $\partial u_j/\partial t_i=-t_j<0$).

(e) Verification. `check_A1.py` checks the best reply and the factorization (1) with sympy, runs the interval recursion for 40 rounds in exact arithmetic (the table, width $5/2^{k-1}$ and the midpoint formula at every round), checks the payoffs 100/9 and 12.5 and the joint optimum 2.5, and, as a second route, runs brute-force IESDS on the grid $\{0,\frac{1}{6},\dots,10\}$, which leaves exactly $t_1=t_2=10/3$. Tadelis gives no answer.

(f) Common mistakes.

- Answering (b) with $[0,\infty)$ minus nothing, or with $[0,10]$. The bound comes from the best replies, which never exceed 5, not from where the payoff is positive.
- Writing $L_{k+1}=5-L_k/2$: the *upper* bound of the opponent gives the *lower* bound of the best reply, because BR_i is decreasing.
- Claiming that a t outside the best-reply range is "dominated" without naming the dominating strategy. Lemma (1) supplies it: the nearer endpoint.

Answer. (a) $BR_i(t_j)=\max\{0,5-t_j/2\}$. (b) $S_i^1=[0,5]$: any $t_i>5$ is strictly dominated by 5. (c) Rounds give $[2.5,5]$, $[2.5,3.75]$, $[3.125,3.75]$, $[3.125,3.4375]$, …, with widths $5/2^{k-1}$; only $t_1=t_2=10/3$ survives IESDS.

Item 9 (item 9 of the combined list)**: MWG Exercise 8.B.4** Advanced · difficulty 4/5

Source: MWG Exercise 8.B.4, page 262. The statement is reproduced verbatim in the list (item 9); it is not repeated here.

(a) What is asked. Prove that, in iterated deletion of strictly dominated strategies, the set of surviving strategies does not depend on the order in which dominated strategies are deleted.

(b) Method. Strict dominance and its iterated deletion (MWG §8.B, Defs. 8.B.2–8.B.4 and the iterated procedure, pp. 236–241; Lecture 1, slides 23 and 29; Tadelis §4.2, pp. 63–65). Two facts carry the proof: (i) dominance is *inherited* when the opponents' strategy sets shrink; (ii) in a finite game, a dominated strategy has a dominator that is itself undominated, and such a dominator is never deleted. The game is finite, as in MWG §8.B (finite S_i); a final remark shows why finiteness is needed.

(c) Solution. Setting. $\Gamma = [I, \{S_i\}, \{u_i\}]$ with every S_i finite. For a product set $T = \prod_i T_i \subseteq S$, say that $s_i \in T_i$ is *strictly dominated in T by $t_i \in S_i$* if $u_i(t_i, s_{-i}) > u_i(s_i, s_{-i})$ for every $s_{-i} \in T_{-i}$; it is *dominated in T* if this holds for some $t_i \in T_i$.

A *deletion sequence* is a chain $S = T^0 \supseteq T^1 \supseteq \cdots \supseteq T^K$ of product sets in which each T^{k+1} is obtained from T^k by deleting a non-empty set of strategies (of one or several players) each of which is dominated in T^k, and which stops only when T^K contains no dominated strategy. "Order of deletion" means the choice of which dominated strategies to delete at each step; the lecture's procedure (Lecture 1, slide 29) is the sequence that deletes all of them at once. Since S is finite and each step deletes something, every deletion sequence ends after finitely many steps. Claim: all deletion sequences end at the same T^K.

Lemma 1 (inheritance). If $X \subseteq Y$ (that is, $X_j \subseteq Y_j$ for all j), $s_i \in X_i$, and s_i is strictly dominated by t_i in Y, then s_i is strictly dominated by t_i in X.

Proof. The inequality $u_i(t_i, s_{-i}) > u_i(s_i, s_{-i})$ holds for every $s_{-i} \in Y_{-i}$, hence for every $s_{-i} \in X_{-i} \subseteq Y_{-i}$. □

Lemma 2 (an undominated dominator). Fix a product set X and a player i. If $s_i \in X_i$ is strictly dominated in X by some strategy in S_i, then there is a $t_i^* \in S_i$ that strictly dominates s_i in X and is itself not strictly dominated in X by any strategy in S_i.

Proof. Let $D = \{t \in S_i : t \text{ strictly dominates } s_i \text{ in } X\}$, non-empty and finite. "r strictly dominates t in X" is irreflexive and transitive (if $u_i(r, \cdot) > u_i(t, \cdot)$ and $u_i(t, \cdot) > u_i(s_i, \cdot)$ on X_{-i}, then $u_i(r, \cdot) > u_i(s_i, \cdot)$ there), so it is a strict partial order, and the finite set D has an element t_i^* not dominated by any other element of D. If some $r \in S_i$ strictly dominated t_i^* in X, then by transitivity r would dominate s_i, so $r \in D$, a contradiction. □

Lemma 3 (the end point of any sequence is contained in every set of any other sequence). Let $(T^k)_{k=0}^K$ and $(R^l)_{l=0}^L$ be two deletion sequences, with end points $T = T^K$ and $R = R^L$. Then $T \subseteq R^l$ for every l; in particular $T \subseteq R$.

Proof, by induction on l. For $l = 0$, $T \subseteq S = R^0$. Suppose $T \subseteq R^l$, and suppose, for a contradiction, that some $s_i \in T_i$ is deleted at step $l+1$ of the second sequence. Then s_i is strictly dominated in R^l by some $t_i \in R_i^l$. Since $T \subseteq R^l$, Lemma 1 shows that s_i is strictly dominated in T by t_i. By Lemma 2 (with $X = T$) there is a $t_i^* \in S_i$ that strictly dominates s_i in T and is not strictly dominated in T by any strategy in S_i.

Now $t_i^* \in T_i$. Indeed, if t_i^* had been deleted at some step $k+1$ of the first sequence, it would be strictly dominated in T^k by some $r \in T_i^k$; since $T \subseteq T^k$, Lemma 1 makes r strictly dominate t_i^* in

T, contradicting the choice of t_i^*.

So $s_i \in T_i$ is strictly dominated in T by $t_i^* \in T_i$: s_i is dominated in T. But T is the end point of a deletion sequence and contains no dominated strategy. This contradiction shows that no strategy of T is deleted at step $l+1$, so $T \subseteq R^{l+1}$. □

The theorem. Apply Lemma 3 twice: $T \subseteq R$, and, exchanging the roles of the two sequences, $R \subseteq T$. Hence $T = R$: every order of deletion ends at the same set, which is the set S^∞ produced by deleting everything dominated in each round. ■

(No T_i^k ever becomes empty: dominance in T^k is a strict partial order on the finite set T_i^k, so T_i^k has an element that no other element of T_i^k dominates, and that element is not deleted at step $k+1$.)

Why finiteness matters. Take one player with $S = [0,1]$ and $u(s) = s$ for $s < 1$, $u(1) = 0$. Every $s < 1$ is strictly dominated by $(1+s)/2$, and $s = 1$ by $s = \frac{1}{2}$. Deleting everything dominated at once leaves the empty set. Deleting only the interval $(0,1)$ first leaves $\{0,1\}$, where $u(0) = u(1) = 0$ and nothing is dominated, so the process stops at $\{0,1\}$. Lemma 2 is what fails: the set of dominators of a strategy has no undominated element.

Why "strictly" matters (the manual's note, made concrete).

$1 \setminus 2$	L	R
T	1, 1	0, 0
M	1, 1	2, 1
B	0, 0	2, 1

With weak dominance: deleting T (weakly dominated by M) first makes L weakly dominated by R on rows $\{M, B\}$, and the process ends at $\{M, B\} \times \{R\}$ with payoffs $(2,1)$. Deleting B (weakly dominated by M) first makes R weakly dominated by L on rows $\{T, M\}$, and it ends at $\{T, M\} \times \{L\}$ with payoffs $(1,1)$. Lemma 1 fails for weak dominance: removing opponent strategies can remove the only profile where the inequality was strict.

(d) Interpretation. IESDS is a well-defined solution concept: what common knowledge of rationality rules out does not depend on the order in which an analyst (or the players) notices dominated strategies. That is why the slides can delete everything at once and why textbook solutions that delete one strategy at a time reach the same answer. Weak dominance does not have this property, which is one reason the course does not iterate it.

(e) Verification. `check_A1.py` draws 2,000 random finite games (two or three players, up to four strategies each, small integer payoffs so that ties are common), runs 20 random deletion orders in each (at each step a random non-empty subset of the currently dominated strategies is deleted) and checks that all of them end at the maximal-deletion set; it also checks the weak-dominance example (two orders, two different end points) and the one-player continuum example on a grid of exact rationals. *Comparison with the MWG solutions manual* (pp. 8-2–8-3): it gives the same key idea in two lines (a dominated s_i has a dominator that is not itself dominated; if s_i is not deleted now it is still dominated, and deleted, later) and the same remark that order matters for weak dominance. It does not show that the undominated dominator survives every later round, nor where finiteness enters; Lemmas 2–3 supply both. The conclusion agrees.

(f) Common mistakes.

- Arguing "a dominated strategy stays dominated, so it is deleted eventually" without checking

that its dominator is still there. The dominator may be deleted; Lemma 2 picks one that cannot be.

- Proving only that the maximal procedure deletes more at each round. The claim is about the end points of *every* order, including orders that delete one strategy at a time.
- Extending the result to weak dominance, or to infinite games without further assumptions; both counterexamples above show it fails.

Answer. In a finite game, strict dominance is inherited by smaller games (Lemma 1), and every dominated strategy has a dominator that is itself undominated (Lemma 2), which no deletion sequence ever removes. So the end point T of any deletion sequence is contained in every stage of any other sequence (Lemma 3), hence in its end point R; by symmetry $R \subseteq T$. All orders give the same surviving set S^∞. (False for weak dominance and, in general, for infinite games: examples above.)

Item 10 (item 10 of the combined list)**: MWG Exercise 8.B.5** Advanced · difficulty 5/5

Source: MWG Exercise 8.B.5, page 262. The statement is reproduced verbatim in the list (item 10); it is not repeated here.

(a) What is asked. Show that IESDS gives a unique prediction in linear Cournot with two firms, and say whether it still does with three.

(b) Method. Iterated deletion of strictly dominated strategies (MWG §8.B, Defs. 8.B.2–8.B.4 and the iterated procedure, pp. 236–241; Lecture 1, slide 29), applied to a continuum of quantities by bounding best replies, as in Tadelis §4.2.2, pp. 65–67, and the slides' Cournot remark (Lecture 2, slide 24: "eliminating outputs that are never optimal yields the sequence $\frac{1}{2}q^m, \frac{3}{4}q^m, \frac{5}{8}q^m, \cdots \to \frac{2}{3}q^m$"). The tool is the dominance lemma of the roommates exercise (Tadelis 4.6), here with a slope b. Assumptions: $a > c \geq 0$, $b > 0$, $q_i \in [0, \infty)$; write $m = (a-c)/(2b)$ for the monopoly output.

(c) Solution. Payoff and best reply. With Q_{-i} the rivals' total output,

$$u_i = \big(a - b(q_i + Q_{-i}) - c\big)\, q_i = b\, q_i\Big(\frac{a-c}{b} - Q_{-i} - q_i\Big) = b\, q_i\,(2x - q_i), \qquad x = m - \tfrac{Q_{-i}}{2},$$

since $(a-c)/b - Q_{-i} = 2m - Q_{-i} = 2x$. This is a concave parabola in q_i with peak at x: $\partial u_i/\partial q_i = b(2x - 2q_i)$, $\partial^2 u_i/\partial q_i^2 = -2b < 0$. With $q_i \geq 0$,

$$BR_i(Q_{-i}) = \max\{0,\ x\} = \max\Big\{0,\ m - \frac{Q_{-i}}{2}\Big\}.$$

Expanding, for any two outputs a' and t,

$$u_i(a', Q_{-i}) - u_i(t, Q_{-i}) = b\big[2x(a'-t) - (a'^2 - t^2)\big] = b\,(t - a')(t + a' - 2x). \tag{1}$$

So if every possible peak x lies in $[\underline{a}, \bar{a}]$, outputs above $\bar{a}$ are strictly dominated by $\bar{a}$ (both factors positive) and outputs below $\underline{a}$ by $\underline{a}$ (both negative). An output that is the unique best reply to some surviving Q_{-i} is not dominated.

(a) Two firms. Here $Q_{-i} = q_j$. If $S_j^k = [L_k, U_k]$, the best replies to $q_j \in [L_k, U_k]$ fill $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = \max\{0,\ m - \tfrac{U_k}{2}\}, \qquad U_{k+1} = m - \tfrac{L_k}{2}.$$

- Round 1: $q_j \in [0, \infty)$, so the peaks $x = m - q_j/2$ are all $\leq m$ and the best replies fill $[0, m]$ ($BR = m$ at $q_j = 0$, $BR = 0$ for $q_j \geq 2m$). By (1) every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: $q_j \in [0, m]$ gives peaks in $[m - m/2, m] = [m/2, m]$; every $q_i < m/2$ is dominated by $m/2$. $S_i^2 = [m/2, m]$.
- Round 3: $q_j \in [m/2, m]$ gives $[m - m/2, m - m/4] = [m/2, 3m/4]$. Round 4: $[m - 3m/8, m - m/4] = [5m/8, 3m/4]$. Round 5: $[5m/8, 11m/16]$.

From round 2 on the bounds are interior ($U_k \leq m < 2m$), so subtracting the recursions gives $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$: the width is $m/2^{k-1} \to 0$. Adding them, the midpoint satisfies $M_{k+1} = m - M_k/2$, whose fixed point solves $q = m - q/2$, i.e. $q = 2m/3$, and $M_k - 2m/3 = (-\frac{1}{2})^{k-1}(M_1 - 2m/3) \to 0$. The nested intervals shrink to one point:

$$S_i^\infty = \Big\{\tfrac{2m}{3}\Big\} = \Big\{\frac{a-c}{3b}\Big\}, \qquad i = 1, 2.$$

IESDS predicts the Cournot outcome $q_1 = q_2 = (a-c)/(3b)$, the slides' $\frac{2}{3}q^m$ (Lecture 2, slide 24; Tadelis's example has $m = 45$ and gives 30).

(b) Three firms. Now $Q_{-i} = q_j + q_h$.

- Round 1: $Q_{-i} \in [0, \infty)$, so the best replies fill $[0, m]$ and every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: now $Q_{-i} \in [0, 2m]$, so the peaks are $x = m - Q_{-i}/2 \in [0, m]$ and $BR(2m) = \max\{0, m - m\} = 0$. The best replies still fill all of $[0, m]$: each $q \in [0, m]$ is the unique best reply to $Q_{-i} = 2(m - q)$, which is reached by $q_j = q_h = m - q \in [0, m]$, two strategies that survived round 1. Nothing is deleted: $S_i^2 = S_i^1$.

Since round 2 deletes nothing, the process stops: $S_i^\infty = [0, (a-c)/(2b)]$ for each firm. *No*, IESDS does not give a unique prediction with three firms. The three-firm Cournot equilibrium $q_i = (a-c)/(4b) = m/2$ (the n-firm exercise, Tadelis 5.8, with $n = 3$ and $b = 1$) lies inside, but so does every other output in $[0, m]$.

(d) Interpretation. With two firms the best-reply map is a contraction (slope $-\frac{1}{2}$): each round of "I know that you know ..." halves the range of plausible outputs, so common knowledge of rationality alone pins down the Cournot outcome. With three firms the rivals' *combined* output can be as large as $2m$, enough to make zero output a best reply, so the first-round bounds reproduce themselves and higher-order reasoning adds nothing. IESDS is weak in games with several players whose actions are strategic substitutes; Nash equilibrium (Lecture 2) still pins down the outcome, but only by assuming correct beliefs.

(e) Verification. `check_A1.py` verifies with sympy that the Cournot payoff equals $bq_i(2x - q_i)$ with $x = m - q_j/2$, the factorization (1), and that the fixed point of $q = m - q/2$ is $(a-c)/(3b)$. It runs the duopoly interval recursion for 60 rounds in exact arithmetic ($m = 1$: $[0, 1]$, $[\frac{1}{2}, 1]$, $[\frac{1}{2}, \frac{3}{4}]$, $[\frac{5}{8}, \frac{3}{4}]$, $[\frac{5}{8}, \frac{11}{16}]$, ..., $\to \frac{2}{3}$), checks that every $q \in [0, m]$ is the best reply to $Q = 2(m - q)$ with both rivals in $[0, m]$, and runs brute-force IESDS for three firms on a grid ($a - c = 12$, $b = 1$, steps of 0.5), which leaves exactly $[0, 6] = [0, m]$. *Comparison with the MWG solutions manual* (pp. 8-3–8-4): it agrees on both parts, $(a-c)/(3b)$ for two firms and $q_i \in [0, (a-c)/(2b)]$ with no unique prediction for three.

(f) Common mistakes.

- Stopping (a) at "the best replies cross at $(a-c)/(3b)$". That is the Nash argument; the exercise asks for dominance, so each round must name what is deleted and by what.
- In (b), bounding Q_{-i} by m instead of $2m$. Each rival can produce up to m, so the sum reaches $2m$, and that is what makes $q_i = 0$ a best reply.
- Concluding from (b) that the three-firm game has many Nash equilibria. It has one; only the dominance argument fails to find it.

Answer. (a) $BR_i(q_j) = \max\{0, (a-c)/(2b) - q_j/2\}$; IESDS gives $[0, m]$, $[m/2, m]$, $[m/2, 3m/4]$, $[5m/8, 3m/4]$, ..., with widths halving, so the unique survivor is $q_1 = q_2 = (a-c)/(3b)$. (b) No: with three firms the rivals' total can reach $2m$, every $q_i \in [0, (a-c)/(2b)]$ is a best reply to some surviving profile, and IESDS stops at $[0, (a-c)/(2b)]$ for each firm.